
CEO salary: Inference and Prediction

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Abstract

While most employees earn a steady wage fixed by contracts and labour agreements, CEO pay is generally much more flexible. Because top executives have specialised skills and significant influence over their companies, their salary packages are often designed to change based on how well the business performs. To make sure the interests of the CEO match those of the shareholders, firms typically link executive pay to specific measures of company success.

The main aim of this study is to analyse whether CEO pay is systematically associated with firm performance. Using a cross-sectional dataset of 209 variables from 1990, containing information on firm size, performance indicators, and industry affiliation, the study employs both simple and multiple linear regression models to assess the role of performance in determining executive compensation.

In addition to these parametric methods, non-parametric techniques, including k-nearest neighbours regression and decision trees, are applied to allow for nonlinear effects and more flexible relationships. Moreover, a simulation study is carried out to evaluate and compare the predictive performance of these different methods under different data-generating processes, offering insight into when non-parametric methods may provide advantages over traditional linear models.

1 Introduction and Motivation

1.1 Case description

Compensation schemes for most employees are typically determined by collective labour agreements, resulting in limited wage flexibility and little direct responsiveness to firm-specific performance. CEOs' compensation, in contrast, is considerably more flexible. The CEO role requires a highly specialised skill set and is associated with a limited supply of qualified candidates, which confers substantial bargaining power in the executive labour market. To align managerial incentives with shareholder interests, CEO compensation is therefore commonly tied to one or more measures of firm performance.

1.2 Research Question

This report investigates whether, and to what extent, CEO compensation is determined by firm performance. Using a cross-sectional dataset of 209 observations from 1990, the analysis examines the relationship between CEO salary and various indicators of firm performance, alongside firm size and industry characteristics. The central research question is whether performance measures significantly contribute to explaining CEO compensation, and whether they improve explanatory and predictive performance beyond firm size alone.

1.3 Main Findings

This study examines the determinants of CEO compensation using both parametric and nonparametric methods. Across several approaches, firm size emerges as the dominant factor in explaining and predicting CEO salary. Linear regression results indicate that, in addition to firm size, measures of firm performance contribute only a little to explaining CEO compensation. The KNN approach complements this analysis by improving the predictive performance findings. Besides the findings of using decision trees are in line with our earlier findings, but highlights differences in how CEO compensation is determined for smaller firms compared to larger firms.

1.4 Outlines

The report is structured as follows. The first part presents the methodology, outlining the theoretical framework and empirical techniques used to analyse the data. This section discusses four complementary approaches: linear regression, k-nearest neighbours (KNN) regression, decision trees, and a simulation study, each highlighting different aspects of inference and prediction. The second part presents the empirical results for each method, comparing their explanatory power, predictive performance and underlying assumptions. The final part concludes by summarising the main findings, discussing their implications for CEO compensation design and reflecting on the strengths and limitations of the applied methods.

2 Data

2.1 Data inspection

Before estimating the regression model, it is essential to conduct an exploratory analysis of the data. The dataset consists of 209 observations, each containing information on the following variables: `salary`, `pcsalary`, `sales`, `roe`, `ros`, `pcroe`, `indus`, `finance`, `consprod`, `utility`, `lsalary`, and `lsales`. In this dataset, no missing values or impossible values are found.

The variables `indus`, `finance`, `consprod` and `utility` are dummy variables indicating the industry to which a firm belongs. Firm size is measured using `sales` and `lsales` (the natural logarithm of sales), while firm performance is captured by `roe` (return on equity), `ros` (return on stock), and `pcroe` (percentage change in return on equity). The primary variable of interest is CEO compensation. Three alternative measures are available for this dependent variable: `salary`, `lsalary` (the natural logarithm of salary), and `pcsalary`.

The heatmap in Figure 1 reveals a strong correlation between `lsales` and `sales`, suggesting that only one of these variables should be included in the model to avoid redundancy.

When examining the summary statistics in Table 1, we observe that for several variables, the maximum and minimum values deviate substantially from the mean. This suggests the potential presence of outliers in the dataset. To investigate this further, it is necessary to perform appropriate data pre-processing.

	<code>salary</code>	<code>pcsalary</code>	<code>sales</code>	<code>roe</code>	<code>pcroe</code>	<code>ros</code>	<code>lsalary</code>	<code>lsales</code>
Count	209	209	209	209	209	209	209	209
Mean	1281.12	13.28	6923.79	17.18	10.80	61.80	6.95	8.29
Std. Dev.	1372.35	32.63	10633.27	8.52	97.22	68.18	0.57	1.01
Min	223.00	-61.00	175.20	0.50	-98.90	-58.00	5.41	5.17
25th pct.	736.00	-1.00	2210.30	12.40	-21.20	21.00	6.60	7.70
Median	1039.00	9.00	3705.20	15.50	-3.00	52.00	6.95	8.22
75th pct.	1407.00	20.00	7177.00	20.00	19.50	81.00	7.25	8.88
Max	14822.00	212.00	97649.90	56.30	977.00	418.00	9.60	11.49

Table 1: Descriptive statistics of CEO salary and firm characteristics

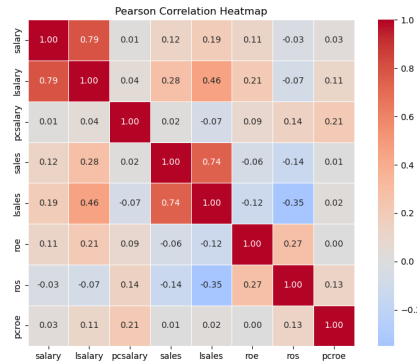


Figure 1: Correlation Heatmap of CEO salary and firm characteristics

2.2 Dealing with outliers

Linear regression, estimated via Ordinary Least Squares, is sensitive to outliers and influential observations, which can have a disproportionate impact on the estimated coefficients. However, this does not imply that every observation lying far from the bulk of the data automatically qualifies as a problematic outlier that must be removed.

When an outlier lies along the overall trend of the regression line, it can still contain useful information. For example, consider a very large firm with a correspondingly high CEO salary. Although this observation is extreme relative to the bulk of the data, it is consistent with the underlying relationship between firm size and compensation. As such, it provides useful information for estimating the regression line and does not need to be removed from the dataset, despite being classified as an outlier.

In contrast, when you consider a firm with exceptionally large size but unusually low CEO compensation, it lies far from the general trend in the data. Because it combines high leverage with a large residual, this observation exerts

a disproportionate influence on the estimated coefficients and can substantially distort the slope of the regression line. In this case, the observation is unlikely to reflect the systematic relationship of interest and may instead arise from measurement error or unobserved firm-specific factors. Consequently, such an outlier requires closer examination and may justify exclusion from the analysis.

To systematically identify potentially problematic outliers, a combination of Cook’s distance, standardised residuals, and leverage statistics is employed. Cook’s distance quantifies the overall influence of an observation on the estimated regression coefficients by incorporating both the size of its residual and its leverage. Standardised residuals highlight observations that deviate substantially from the values predicted by the model, while leverage measures the extent to which an observation has extreme values of the explanatory variables. Observations exhibiting both large residuals and high leverage are considered potentially problematic, as they can disproportionately affect the fitted regression line. This combined approach distinguishes between informative extreme observations and outliers that may distort the estimation of the model.

Rather than relying solely on the removal of extreme observations, logarithmic transformations can also be applied to highly skewed variables such as CEO salary and firm sales. Log transformations reduce the impact of large values, compress the scale of the data and often lead to more symmetric distributions. This approach allows the model to retain economically meaningful variation while reducing the sensitivity of OLS estimation to extreme observations.

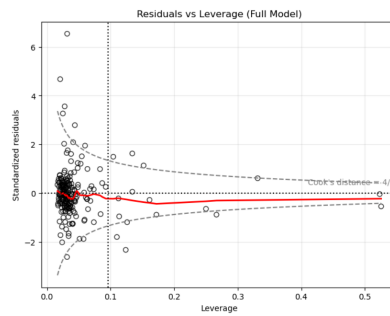


Figure 2: Standardised residuals versus Leverage

As a first step in pre-processing our data, potentially influential observations are identified using standardised residuals, Cook’s distance, and leverage. The results of this diagnostic analysis are presented in Figure 2. The figure indicates the presence of 16 potentially influential observations.

Removing all of these observations would require excluding nearly ten per cent of the sample, which could substantially reduce statistical power and bias the regression results, given the relatively modest sample size. Therefore, instead of deleting observations, an alternative approach is adopted to mitigate the influence of extreme values.

Specifically, the log transformation of salary and sales is used to reduce the impact of outliers, as seen in the summary statistics of Table 1. Visual inspection of the transformed data shows a substantially more stable distribution, indicating that the influence of extreme observations has been considerably reduced. This is visualised in Figure 3. Consequently, the analysis proceeds using $\ln \text{salary}$ as the dependent variable rather than excluding observations, thereby preserving the sample size while improving the robustness of the regression results.

3 Methodology

3.1 Linear Regression

3.1.1 General description

Linear regression is a statistical method used to model the relationship between a dependent variable and one or more independent variables. The objective is to assess how changes in the explanatory variables are reflected in changes in the outcome variable, assuming a linear relationship in the parameters of the model. In its most general form, a linear regression model can be written as displayed in equation (1).

$$y_i = \beta_0 + \beta_1 x_{1,i} + \beta_2 x_{2,i} + \dots + \beta_k x_{k,i} + \epsilon_i \quad (1)$$

Where y_i denotes the dependent variables, $x_{j,i}$ are the explanatory variables, β_j are unknown parameters to be estimated and ϵ_i represents an error term capturing unobserved factors.

In this study, linear regression is used to analyse the determinants of CEO compensation. Ordinary Least Squares (OLS) estimation is employed, which chooses coefficient estimates that minimise the sum of squared residuals.

3.1.2 Inference versus Prediction

An important distinction in linear regression is between statistical inference and prediction, as linear models can be used for both purposes but are evaluated by different criteria depending on the objective.

When the goal is inference, the primary aim is to approximate the true underlying relationship between the explanatory variables and the dependent variable as accurately as possible. The focus lies on estimating coefficients that reflect meaningful economic relationships and on assessing whether these relationships are statistically significant and consistent with theory. In this context, the classical regression assumptions play a central role because they ensure that the estimated standard errors are valid and that statistical inference is meaningful. If assumptions such as constant error variance, independence of the residuals or correct functional form are violated, the estimated coefficients may still be unbiased, but the associated t-statistics and F-statistics can become misleading. This directly affects the reliability of hypothesis tests and confidence intervals, and may also distort goodness-of-fit measures such as the adjusted R^2 . Therefore, verifying that these assumptions are reasonably satisfied is essential when linear regression is used for inferential purposes.

When the goal is prediction, by contrast, the emphasis shifts away from uncovering the true data-generating process and toward accurately forecasting outcomes for new, unseen observations. In this case, the statistical significance of individual coefficients and strict adherence to model assumptions are less important than the model's out-of-sample predictive accuracy. A model may therefore perform well in prediction even if some coefficients are unstable or difficult to interpret from an inferential perspective.

In this study, both perspectives are considered, but the main focus lies on inference. Inference is used to identify which firm characteristics are systematically associated with CEO compensation. Predictive performance is evaluated using cross-validation, which provides an objective measure of out-of-sample accuracy.

3.1.3 Cross Validation

To assess the predictive performance and generalisability of the regression models, k-fold cross-validation is employed. In this procedure, the dataset is randomly partitioned into k approximately equal-sized folds. Each fold is used once as a validation set, while the remaining $k - 1$ folds serve as the training set. The model is fitted on the training folds, and its predictions are evaluated on the validation fold. For each fold, the mean squared error (MSE) is computed as explained in equation (2).

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (2)$$

This process is repeated k times so that every fold is used once for validation. The out-of-sample prediction error for each fold is then averaged to obtain the cross-validated mean squared error (CV-MSE), as visualised in equation (3). This provides a single measure of predictive accuracy.

$$CV-MSE = \frac{1}{K} \sum_{i=1}^k MSE_i \quad (3)$$

An alternative form of cross-validation is Leave-One-Out Cross-Validation (LOOCV), where k is set equal to the number of observations in the dataset. In LOOCV, each observation serves as a single validation case, and the model is trained on all remaining observations. The prediction errors are then averaged across all observations to calculate the overall CV-MSE. LOOCV provides an almost unbiased estimate of out-of-sample performance, as nearly the entire dataset is used for training in each iteration. It can be particularly advantageous for smaller datasets, where retaining as much training information as possible reduces estimation variance and leads to more reliable predictions. However, because each training set contains only a single observation for validation, the standard deviation of the CV-MSE tends to be much higher compared to k-fold cross-validation. (Rebecca C. Steorts, n.d.)

Cross-validation is particularly valuable in this context because it evaluates the model's ability to forecast CEO compensation for unseen data, rather than merely assessing in-sample fit. Comparing CV-MSE across alternative model specifications, including those with additional explanatory variables, nonlinear terms or industry dummies, helps identify the model that provides the best predictive performance.

3.2 K Nearest Neighbors

3.2.1 K Nearest Neighbours and nested cross-validation

K-Nearest Neighbours (k-NN) is a flexible, non-parametric prediction method that does not impose a functional form on the relationship between the outcome and the covariates. Instead of estimating model parameters, k-NN predicts

outcomes using local information from the training data. For a given observation x_0 , the algorithm identifies the k nearest observations in the training sample according to a distance metric, typically the Euclidean distance. In the regression setting, the predicted value is the average outcome among the k neighbours, as visible in equation (4).

$$\hat{y}(x_0) = \frac{1}{k} \sum_{i \in \mathcal{N}_k(x_0)} y_i, \quad (4)$$

while in the classification setting, the predicted class corresponds to the majority class among the k neighbours. The choice of k governs the bias–variance trade-off: small values of k capture local patterns but may overfit, whereas larger values yield smoother predictions at the cost of increased bias. The optimal k is therefore selected using cross-validation.

We apply k-NN in both classification and regression settings, using cross-validation to assess predictive performance and tune k . In the classification analysis, the continuous outcome `lsalary` is transformed into a binary variable indicating whether an observation lies above or below the sample median. We estimate k-NN classifiers for a range of k values and evaluate performance using 5-fold cross-validated accuracy on the training data, selecting the smallest k that achieves the highest accuracy. The final model is then evaluated on a hold-out test set to assess out-of-sample classification performance.

In the regression analysis, k-NN is used to predict `lsalary` directly. To tune the number of neighbours k in a principled way and avoid overfitting, we implement a nested 5-fold cross-validation procedure on the training data. In the inner loop, candidate values of k are evaluated using cross-validated mean squared error (MSE), and the value of k that minimises the average inner-loop MSE is selected. The outer loop is used to repeat this tuning process across different training partitions, ensuring that the choice of k is stable and not driven by a particular split of the data. The final value of k is chosen as the median of the optimal k values selected across the outer folds. The predictive performance of the resulting model is evaluated only once on the hold-out test set, and the test MSE is reported as the measure of true out-of-sample predictive accuracy. (Scikit-Learn, 2025)

3.2.2 Cross-validated MSE versus Test MSE

Although similar, there is a distinct difference between cross-validated MSE and test MSE for KNN. Cross-validated MSE is computed by repeatedly training the model on subsets of the training data and evaluating it on validation folds. It is primarily used for model selection and hyperparameter tuning, as it provides an internal estimate of out-of-sample performance while still relying on the training data. In contrast, test MSE is calculated on a completely independent test sample that is not used in either model estimation or tuning. As a result, test MSE provides the most reliable measure of a model’s predictive performance and its ability to generalise to new, unseen data. While cross-validated MSE is useful for comparing models and selecting optimal parameters, test MSE remains the primary criterion for evaluating true predictive accuracy.

3.3 Decision Tree

3.3.1 Principles of Regression Trees

Following the analysis via K-Nearest Neighbours, we transition to another non-parametric approach: Decision Trees. These are categorised into classification and regression trees. While classification trees predict categorical data, such as "High" vs. "Low" salary groups, regression trees are utilised to predict numerical data. (Complexica, 2024) Given that our objective is to investigate whether firm performance attributes to predicting CEO salary, a regression tree is the appropriate tool. Our goal is to use recursive binary splitting to partition observations into homogeneous groups based on firm characteristics, such as size and performance. While this creates distinct clusters, the regression tree’s purpose is to calculate the average salary within these groups to provide a numeric prediction that minimises the Residual Sum of Squares (RSS).

One of the main advantages of decision trees is their simplicity. They can be visualised graphically and are highly intuitive to interpret. Unlike many parametric models, trees require minimal preprocessing of the data, such as scaling or normalisation. Moreover, they make no assumptions about the underlying distribution of the data, meaning that the standard assumptions required for linear regression do not need to be satisfied. (GeeksforGeeks, 2026)

3.3.2 Pruning Theory and Constraints

Despite their interpretability, decision trees exhibit several limitations, including a propensity for overfitting and high variance. They are extremely sensitive to minor fluctuations in the training data, where a slight shift in the sample can yield significantly different splits. Furthermore, because trees employ a "greedy" splitting algorithm, making the best

choice at each step rather than looking ahead, they may achieve local optima rather than a global optimum. (Geeks-forGeeks, 2026) To mitigate these risks, we applied pruning.

Pruning is a machine learning technique that reduces the size of decision trees by removing sections of the tree that provide little power to classify instances or predict outcomes. By cutting branches that only account for noise or outliers, pruning simplifies the model and improves its ability to generalise to new, unseen data.

We utilised Cost Complexity Pruning, a post-pruning technique that begins by growing a large, unconstrained tree (T_0) and then introduces a tuning parameter (α), which acts as a complexity penalty. The algorithm identifies a subtree ($T \subset T_0$) for each value of α that minimizes the cost function. (James et al., 2023):

3.3.3 Implementation of Standard and Refined Models

We developed a standard decision tree and a refined version to highlight the necessity of data evaluation and illustrate the inherent limitations of the decision tree framework. For the standard models, we utilised GridSearchCV, a machine learning process that trains different models on the dataset and selects the one with the highest predictive performance. (Great Learning, 2024) GridSearchCV operates by defining a dictionary of potential values for α and training a separate model for every possible value in that grid. Each model is then evaluated using 5-fold cross-validation, which calculates the average predictive performance across different subsets of the data. This process allows us to automate the tuning of hyperparameters and helps us select the model with the best performance.

For the refined version, we implemented an explicit shuffling mechanism within a KFold strategy. Shuffling is a critical step in cross-validation to ensure that the distribution of the target variable and features is consistent across all folds. Without shuffling, if a dataset contains any inherent ordering or clustering, the cross-validation folds may be dominated by specific patterns, leading to skewed performance estimates and biased results. Furthermore, we incorporated pre-pruning by implementing a constraint requiring a minimum amount of data for each split. Unlike cost-complexity pruning, which simplifies a tree after it is fully grown, pre-pruning acts as a "stop-growth" rule that prevents the tree from creating splits that are too specific. (Mohan, 2020) By requiring a representative number of observations in each node, we ensured that each prediction is based on a significant group rather than an outlier, thereby improving the model's stability and reducing the risk of capturing stochastic noise.

3.3.4 Ensemble Methods: Random Forest

The final stage of our analysis involved the implementation of a Random Forest regressor, an ensemble learning technique that aggregates the predictions of 500 individual decision trees. While our refined trees provided a clear, interpretable flowchart of compensation logic, the Random Forest was introduced with the goal of helping to improve accuracy and reduce errors. By combining multiple trees, the model effectively addresses the high variance issues discussed previously, providing a more robust final prediction for CEO compensation. (GeeksforGeeks, 2025)

3.4 Simulation Study

Now we have finished addressing the main research question. The only remaining part is the simulation study. The goal of this simulation study is to compare the predictive performance of linear regression, K-nearest neighbours (KNN), and decision trees. This allows us to assess how well each method can be trusted relative to the others.

To make the simulation realistic, we designed the data-generating process (DGP), as displayed in equation (5), to closely resemble the relationships found in our dataset. Specifically, we included a nonlinear effect using a squared term and generated non-normal errors with a Student-t distribution with three degrees of freedom to reflect the heavy-tailed nature of our residuals. We also included dummy variables for categorical predictors, which were simulated using a binomial distribution. We simulated the continuous variables from normal distributions. Importantly, we used different parameters for these normal distributions to reflect the fact that the continuous variables in our dataset are on different scales. For example, $lsales$ was simulated with a mean of 8.3 and standard deviation of 1.0, while roe was simulated with a mean of 17.2 and standard deviation of 8.5, as can be seen from Table 1. This ensures the simulation captures the small-scale differences present in our actual data.

We generated a small dataset of exactly 209 observations to closely resemble the dataset used in this report, allowing for a more meaningful comparison of the predictive performance of the different methods. The coefficients were chosen based on the results from our own linear regression to best reflect the relationships in the original dataset.

$$y_i = 1.5 + 0.25 lsales_{c,i} - 0.05 lsales_{c,i}^2 + 0.005 roe_i + 0.1 finance_i + 0.35 consprod_i - 0.3 utility_i + \varepsilon_i \quad (5)$$

Because our dataset was relatively small, we naturally faced the question of whether we should remove the outliers we identified. To investigate how the presence or removal of outliers affects the performance of the three methods, we designed three simulation scenarios:

1. **No outliers:** In the first scenario, no outliers are present, allowing us to examine the effect of removing all structural outliers on model performance.
2. **Mild outliers (5%):** The second scenario contains only 5% mild outliers, to test the effect of small outliers on the models. This scenario is comparable to how our real dataset looked before applying a log-transformation.
3. **Extreme outliers (15%):** In the final scenario, 15% of the observations are extreme outliers, simulating a worst-case situation to see which model is most sensitive to a dataset containing a lot of extreme values.

This setup allows us to systematically assess the robustness of each method to outliers of varying severity. For testing which model performs best, the mean squared error (MSE) is used, as presented in equation (2). To assess the variability and reliability of the predictive performance, we conducted a Monte Carlo simulation with 500 repetitions. In each repetition, a new dataset is generated based on our data-generating process, the models (linear regression, K-nearest neighbours, and decision trees) are trained and tested, and the test MSE is recorded. For the K-nearest neighbours model, nested K-fold cross-validation is used to select the optimal number of neighbours k , and both scaled and unscaled versions of the predictors are tested to examine the effect of the small differences in scale among the continuous variables. For the decision tree model, K-fold cross-validation is applied for cost-complexity pruning to select the optimal tree size.

In summary, this simulation study provides a controlled framework to compare the predictive performance and robustness of linear regression, K-nearest neighbours and decision trees under different outlier scenarios. The results of these simulations, including average test MSE and box plots of our results, are presented and discussed in the Results section below.

4 Results

4.1 Linear Regression

4.1.1 Simple Linear Regression

To initiate the linear regression analysis, a simple linear regression model is estimated as a baseline for the more comprehensive models introduced later. Following the basic regression specification presented in equation (1), the model is estimated with $k = 1$, where `lsalary` is the dependent variable and `lsales` is the sole explanatory variable. Firm size is expected to play an important role in determining CEO compensation, which motivates the inclusion of `lsales` as the regressor. The estimation results are visualised in Figure 3.

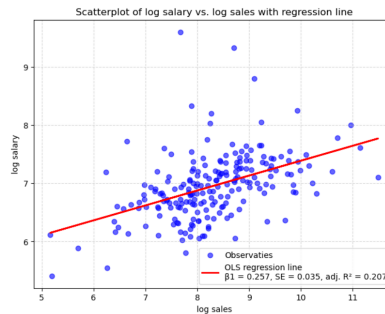


Figure 3: Simple linear regression model: `lsalary` vs. `lsales`

The estimated slope coefficient is positive, with a value of $\beta_1 = 0.257$, indicating a positive relationship between firm size and CEO compensation. In the context of a log-log model, this coefficient implies that a one per cent increase in firm sales is associated with an approximate 0.257 per cent increase in CEO salary, on average. This result aligns with economic intuition, as larger firms tend to exhibit greater organisational complexity and managerial responsibility, which is reflected in higher executive compensation. The adjusted- R^2 tells us that the model explains approximately 20% of the variance in CEO compensation. The intercept has no meaningful economic interpretation, as it corresponds to a firm size of zero, which is outside the range of observed data.

This baseline regression provides empirical support for the relevance of firm size in explaining CEO compensation and serves as a reference point for the extended regression models discussed in subsequent sections.

4.1.2 Extending with Performance measures

After establishing a baseline relationship between CEO salary and firm size through the simple linear regression model, the analysis is extended by incorporating additional explanatory variables that capture firm performance. While the sim-

ple model provides an initial benchmark, it is unlikely to fully explain variation in CEO compensation, as compensation schemes are typically designed to reward not only firm size but also profitability and operational efficiency.

In addition, the following performance measures are added: `roe` (return on equity), `ros` (return on sales) and `pcroe` (percentage changes in return on equity). Following the basic regression specification presented in equation (1), the model is estimated with $k = 4$, where `lsalary` is the dependent variable and `lsales`, `roe`, `ros` and `pcroe` are the regressors.

The estimation results indicate that the coefficients on `pcroe` and `ros` are statistically insignificant at conventional levels (p-values of 0.119 and 0.835, respectively), as displayed in Table 2. Statistical insignificance implies that, given the available data, there is insufficient evidence to conclude that these variables have a systematic effect on CEO compensation. In practical terms, this means that variations in `pcroe` and `ros` do not reliably explain differences in logarithmic CEO salary in this sample. Including variables that do not contribute meaningful explanatory power can increase model complexity without improving predictive accuracy, and may even introduce noise that reduces the precision of other coefficient estimates.

Consequently, it is reasonable to consider a model with $k = 2$, where `lsalary` is the dependent variable and only `lsales` and `roe` as regressors. This slightly reduces the adjusted- R^2 , as visible in Table 2. However, it ensures that all included variables are statistically significant. Moreover, the simplified model enhances interpretability by reducing complexity, potentially lowering variance at the cost of a modest increase in bias.

	Coefficient	p-value
Intercept	4.3622	0.000
lsales	0.2751	0.000
roe	0.0179	0.000
Adjusted R^2	0.275	
Observations	209	

$$lsalary \sim lsales + roe$$

	Coefficient	p-value
Intercept	4.344	0.000
lsales	0.276	0.000
roe	0.018	0.000
pcroe	0.001	0.119
ros	0.0001	0.835
Adjusted R^2	0.277	
Observations	209	

$$lsalary \sim lsales + roe + pcroe + ros$$

Table 2: Comparison of regression specifications

Compared to the simple linear regression model with only `lsales` as a regressor, the inclusion of a performance measure, with as best option the inclusion of `roe`, improves the explanatory power of the model. This is visible from the increased adjusted- R^2 . It therefore provides a more complete understanding of the determinants of CEO compensation. Therefore, it can be concluded that performance measure helps to determine CEO salary.

4.1.3 Non-linear relationships

For explaining CEO compensation, it is possible that diminishing marginal returns exist. This indicates that, for larger firms, increases in firm size or performance have a smaller effect on CEO compensation compared with smaller firms. To examine potential signs of functional misspecification, the residuals versus fitted values were plotted for the model described in Section 4.1.2, as shown in Figure 4a. A LOWESS trend line in a residuals versus fitted values plot illustrates the smoothed relationship between residuals and predicted values, facilitating the detection of systematic patterns. If the trend line deviates noticeably from zero, it suggests functional misspecification, indicating that the chosen regression model may not fully capture the true relationship between the variables.

In Figure 4a, a slight U-shaped pattern is visible in the residuals, suggesting the presence of functional form misspecification. To address this issue, a model with $k = 3$, including the regressors `lsales`, `lsales2` and `roe`, is estimated. As shown in the residuals versus fitted values plot in Figure 4b, the U-shaped pattern is no longer present, providing initial evidence that the functional form has been appropriately adjusted. To formally evaluate this improvement, a RESET test was conducted, as described in Section 6.0.1. The results, presented in Table 3, confirm that the inclusion of the squared term effectively resolves the functional misspecification.

However, when examining Table 4, a new issue arises: the newly added squared term exhibits high multicollinearity with the existing `lsales` variable, as evidenced by the Variance Inflation Factor (VIF) test discussed in Section 6.0.1. High multicollinearity can inflate standard errors and reduce the precision of coefficient estimates, making it difficult to disentangle the individual effects of correlated regressors. One effective approach to mitigate this problem, without altering the underlying data, is to centre the variables by subtracting their means. By creating a centred version of

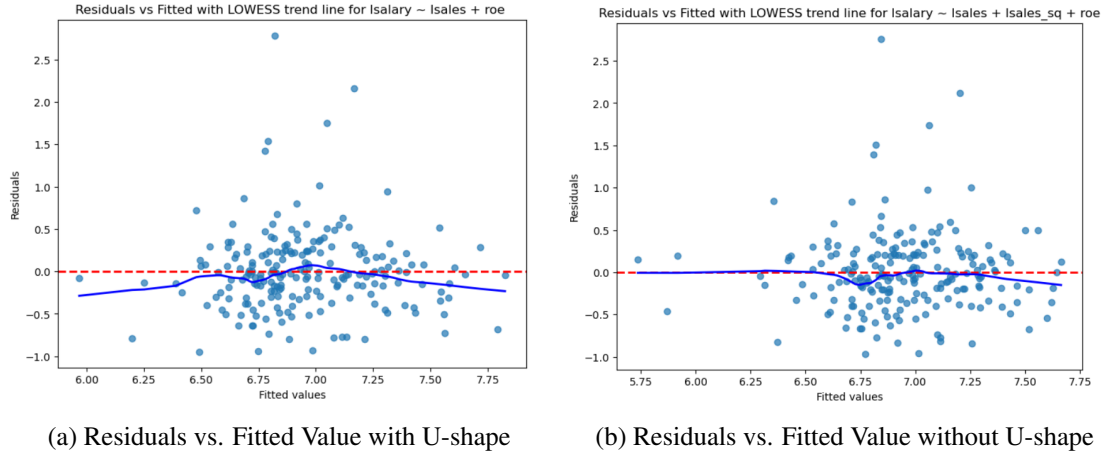


Figure 4: Residuals vs. fitted values before and after adding a squared term

$lsales_c$ and its squared term, the correlation between the two regressors is substantially reduced, resulting in much lower VIF values and improving the stability and interpretability of the regression coefficients.

After extending the existing regression model with the demeaned squared term, $lsales_c^2$, the results indicate the presence of diminishing marginal returns: the coefficient for $lsales_c$ is positive, while the coefficient for its squared term is negative, as visible in Table 5.

Model	p-value
$lsalary \sim lsales + roe$	0.025
$lsalary \sim lsales + lsales^2 + roe$	0.347

Table 3: RESET Test Results

Variable	VIF
const	1741.27
lsales	100.84
lsales_sq	100.66
roe	1.02

VIF with squared term

Variable	VIF
const	5.37
lsales_c	1.02
lsales_c_sq	1.01
roe	1.02

VIF after centering the data

Table 4: Variance Inflation Factors (VIF) before and after centering the data.

4.1.4 Qualitative variables

The model is now further extended by incorporating qualitative information through industry dummy variables. The dataset contains four industry categories: *finance*, *indus* (industrial), *consprod* (consumer products), and *utility*. Including dummy variables allows the model to account for systematic differences in CEO compensation across industries, which may reflect sector-specific practices, market conditions, or compensation norms.

However, when adding dummy variables, care must be taken to avoid the dummy variable trap. This occurs when all categories are included simultaneously, which introduces perfect multicollinearity: the sum of the dummies would equal one, making one variable a perfect linear combination of the others. To prevent this, we include only $k - 1$ dummy variables for a categorical variable with k categories. In our model, we include *finance*, *consprod*, and *utility* as dummies, while the industrial sector (*indus*) serves as the reference category. The intercept of the regression then represents the baseline effect for the reference group, and the coefficients of the included dummies measure the difference in CEO compensation relative to this baseline.

Following the basic regression specification presented in equation (1), the model is extended to $k = 6$, with regressors $lsales_c$, $lsales_c^2$, *roe*, *finance*, *consprod* and *utility*. A joint F-test indicates that the set of industry dummies is jointly statistically significant, implying that industry affiliation contributes to explaining variation in CEO compensation. This, combined with the fact that performance measures such as the adjusted- R^2 increase substantially, suggests that the inclusion of these variables improves the explanatory power of the model.

Although the coefficient on the `finance` dummy is not individually statistically significant, as indicated in Table 5. Its inclusion remains important for correct model specification. Omitting this dummy would alter the composition of the reference category by absorbing observations from the finance industry, thereby changing the interpretation of the remaining dummy coefficients and potentially biasing comparisons across industries. Retaining all relevant industry indicators ensures that each sector is appropriately represented and that the estimated coefficients are interpreted relative to a consistent baseline category.

Additionally, it is plausible that CEO compensation depends less on the exact magnitude of firm performance and more on whether performance is negative, measured through the return on stock. To investigate this, a dummy variable is constructed, taking the value one when `ros` is negative and zero otherwise. While `ros` was previously not statistically significant, the direction of stock returns may nonetheless provide relevant information about CEO salary. When this qualitative performance indicator is included in the regression, the continuous performance measure `roe` becomes statistically insignificant, indicating that its explanatory power is largely captured by the indicator reflecting negative performance.

Variable	Coeff.	p-val	Variable	Coeff.	p-val	Variable	Coeff.	p-val
Intercept	7.041	0.000	Intercept	6.804	0.000	Intercept	6.988	0.000
<code>lsales_c</code>	0.267	0.000	<code>lsales_c</code>	0.260	0.000	<code>lsales_c</code>	0.249	0.000
<code>lsales_c²</code>	-0.057	0.003	<code>lsales_c²</code>	-0.055	0.005	<code>lsales_c²</code>	-0.059	0.003
NegROS	-0.315	0.003	<code>roe</code>	0.010	0.016	<code>finance</code>	0.086	0.332
<code>finance</code>	0.072	0.409	<code>finance</code>	0.120	0.178	<code>consprod</code>	0.257	0.002
<code>consprod</code>	0.223	0.006	<code>consprod</code>	0.203	0.016	<code>utility</code>	-0.407	0.000
<code>utility</code>	-0.447	0.000	<code>utility</code>	-0.339	0.001	Adj. R^2	0.348	
Adj. R^2	0.374		Adj. R^2	0.364		Obs.	209	
Obs.	209		Obs.	209		Model 3		
Model 1			Model 2					

Table 5: Comparison of regression models explaining CEO compensation

Following the basic regression specification presented in equation (1), the model has still $k = 6$, but now the regressors are `lsalesc`, `lsalesc2`, NegROS, `finance`, `consprod` and `utility`. A comparison of the two specifications shows that both models perform almost similarly in explaining CEO compensation. However, the adjusted- R^2 is slightly higher for the model incorporating NegROS, indicating a marginal improvement in explanatory power despite the use of less detailed performance information. In comparison with Model 3, which only includes firm size and industry categories, we see a slight improvement in the explanatory power of the model by adding performance measures. However, this increase is modest and not as large as when firm size or dummy variables were added. Across all models, the coefficients on firm size are large and very stable, indicating that firm size is a dominant determinant of CEO pay. This suggests that firm size and industry affiliation account for a substantially larger share of the variation in CEO compensation than firm performance measures, which provide only limited additional explanatory power.

But still, adding firm performance slightly improves the model. Although NegROS contains less detailed information than a continuous performance measure, it captures a qualitatively important threshold effect in CEO compensation. Executive pay contracts often penalise poor performance more strongly than they reward incremental improvements. As a result, whether firm performance is negative or not may matter more for compensation decisions than the exact magnitude of performance.

This result suggests that in linear regression, CEO compensation may respond more strongly to qualitative performance outcomes, such as the occurrence of negative returns, than to marginal changes in continuous performance measures. However, the difference in adjusted- R^2 between the two specifications is very small, indicating that both models perform comparably well in explaining CEO compensation. Given the similarity in explanatory power, neither specification clearly dominates the other, and both provide equally plausible representations of the relationship between firm size, performance, industry affiliation and CEO compensation.

4.1.5 Model assumptions

After evaluating the models for both simplicity and explanatory power using inference, two specifications stand out: Model 1 and Model 2 from Table 5. As outlined in the Appendix Section 6.0.1, inference on linear regression relies on several key assumptions, which were assessed using the tests summarised in Table 13.

Test	Model 1 (NegROS)	Model 2 (ROE)
RESET Test (Functional Form)	0.527	0.049
Breusch-Pagan (Heteroskedasticity)	0.367	0.360
Jarque-Bera (Normality)	0.000	0.000
Durbin-Watson (Autocorrelation)	2.152	2.245

Table 6: Diagnostic p=values for different Tests

Both models satisfy the assumptions of homoskedasticity and independent residuals, and the variance inflation factors indicate no problematic multicollinearity. The RESET test indicates that functional form misspecification is not a concern for Model 1 (NegROS), whereas for Model 2 (ROE) it is marginally significant. This effect arises primarily after including the industry dummy variables in combination with continuous performance measures, suggesting minor non-linear patterns may exist in certain sectors. However, the magnitude of this effect is very small, and it does not meaningfully impact the interpretation or reliability of the model. It is possible, however, that this slight functional misspecification could increase the variance of the coefficient estimates, which in turn might lead to slightly less precise predictions compared with the simpler NegROS specification.

The Jarque-Bera test indicates deviations from normality in the residuals for both models, which may partly result from the relatively small sample size, a known limitation that can amplify skewness and kurtosis. This can bias standard errors and confidence intervals. Despite these minor issues, the non-normal errors do not undermine the primary goal of estimating unbiased coefficients, and overall, the diagnostics support the reliability of both models for interpretation and prediction.

4.1.6 Cross validation

Model selection was initially conducted based on statistical inference, aiming to identify the specification that best captures the underlying determinants of CEO compensation. To evaluate predictive performance, however, two types of cross-validation, as described in Section 3.1.3, were applied to the models identified as most prominent. The selected models and their specifications are summarised in Table 7.

Model	Specification
1	$lsalary \sim lsales_c + lsales_c^2 + finance + consprod + utility$
2	$lsalary \sim lsales_c + lsales_c^2 + roe + pcroe + ros + finance + consprod + utility$
3	$lsalary \sim lsales_c + lsales_c^2 + roe + finance + consprod + utility$
4	$lsalary \sim lsales_c + lsales_c^2 + roe + ros + finance + consprod + utility$
5	$lsalary \sim lsales_c + lsales_c^2 + roe + NegROS + finance + consprod + utility$
6	$lsalary \sim lsales_c + lsales_c^2 + roe + pcroe + NegROS + finance + consprod + utility$
7	$lsalary \sim lsales_c + lsales_c^2 + NegROS + finance + consprod + utility$
8	$lsalary \sim roe + NegROS + pcroe + finance + consprod + utility$

Table 7: Regression model specifications for CEO compensation

When examining the cross-validation results, we observe that the models combining firm size with industry dummy variables and the NegROS indicator consistently perform best in terms of predictive accuracy. In these specifications, NegROS may be complemented by other measures of firm performance. However, the results clearly show that this simple indicator, capturing whether a firm makes a profit or incurs a loss, remains crucial. This suggests that, for predicting CEO compensation, the distinction between positive and negative performance is more informative than the exact magnitude of performance. In other words, it appears to matter more whether a firm is profitable at all, rather than how profitable it is, when determining CEO salary.

Nevertheless, when comparing these best-performing models to specifications that rely solely on firm size and industry dummies, the differences in predictive performance are remarkably small (≈ 0.01). The model that excludes firm performance variables altogether performs almost equally well. This indicates that firm size and industry categories are the dominant factors in explaining CEO compensation, and that performance measures, when captured in the dummy variable NegROS, only provide a modest incremental improvement. From a predictive standpoint, firm size alone already captures most of the relevant variation in CEO pay.

This finding is economically intuitive. Larger firms typically have greater financial resources and higher organisational complexity, which allows them to offer higher compensation packages to their CEOs. As a result, CEO salary appears to depend primarily on the scale of the firm rather than on short-term performance fluctuations.

The slight decline, or at least the absence of any improvement, in predictive accuracy when only continuous performance measures are included in the baseline model may be due to mild functional misspecification. In particular, if the effect of firm performance on CEO pay is nonlinear or asymmetric, a linear model using continuous measures may not capture this relationship well.

What is clearly visible is that models relying solely on firm performance measures alongside categorical dummies perform substantially worse in prediction, indicating that performance alone is insufficient to explain CEO compensation. Simple performance indicators such as *NegROS* are most effective when complementing firm size and industry controls rather than substituting for them, underscoring that firm size and category dummies remain the primary drivers of predictive accuracy while performance plays a secondary role.

Model	CV MSE ($\pm$ std)	Model	CV MSE ($\pm$ std)
7	0.1872 ($\pm$ 0.0521)	6	0.1792 ($\pm$ 0.4392)
5	0.1929 ($\pm$ 0.0629)	5	0.1793 ($\pm$ 0.4398)
6	0.1936 ($\pm$ 0.0593)	7	0.1813 ($\pm$ 0.4338)
1	0.1975 ($\pm$ 0.0457)	3	0.1847 ($\pm$ 0.4533)
2	0.1994 ($\pm$ 0.0592)	2	0.1857 ($\pm$ 0.4605)
4	0.1995 ($\pm$ 0.0632)	4	0.1865 ($\pm$ 0.4626)
3	0.2002 ($\pm$ 0.0592)	1	0.1898 ($\pm$ 0.4486)
8	0.2561 ($\pm$ 0.0741)	8	0.2499 ($\pm$ 0.5698)

K-fold Cross-Validation (test size = 0.2)

LOOCV

Table 8: Comparison of cross-validation results for selected regression models

4.2 K Nearest Neighbours

While cross-validation results for linear regression show that firm size alone already captures most of the predictive information for CEO compensation, and adding performance indicators like *NegROS* yields only minimal gains, this raises the question of whether more flexible, nonparametric methods can better exploit the predictive content of this performance. In particular, linear regression assumes a global functional form and averages the effects of performance across all firms, which may mask local patterns and heterogeneity. To explore this possibility, we apply the k-Nearest Neighbours algorithm, which can capture nonlinear and locally varying relationships between firm size, performance, and CEO pay, and compare its predictive performance to linear regression in both regression and classification settings.

4.2.1 KNN in classification setting

Firstly, the k-Nearest Neighbours algorithm was applied to predict whether an observation falls above or below the median log-salary. We applied this process through 2 models, one only used firm size to predict CEO salary, and one with other firms' performance and the type of firms as well. The results are as follows:

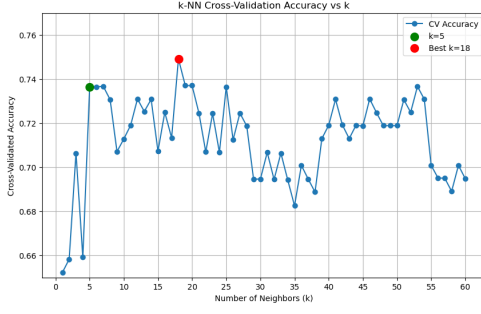
Model	Specification
1	$lsalary \sim lsales_c + lsales_c^2 + finance + consprod + utility$
2	$lsalary \sim lsales_c + lsales_c^2 + NegROS + finance + consprod + utility$

Table 9: Regression model specifications for CEO compensation

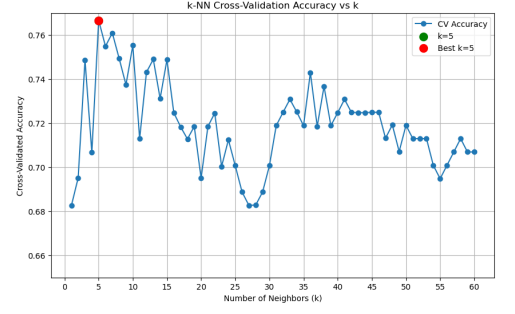
	Model 1	Model 2
Cross_validated Accuracy (k=5)	0.74	0.77
Cross_validated Accuracy (best k)	0.75	0.77
Test Accuracy (k=5)	0.67	0.76
Test Accuracy (best k)	0.74	0.76
Best k	18	5

Table 10: Comparison of test accuracies across 2 different models using KNN cross validation

Looking at table 10, Model 1, which relies only on firm size and firm types, achieves reasonably strong cross-validated accuracy, indicating that size provides meaningful predictive power in classifying CEOs as earning above or below the



Model 1



Model 2

Figure 5: Comparison of cross-validated accuracies across two different models using KNN cross-validation.

median salary. However, looking at figure 5, Model 2, which additionally incorporates firm performance, consistently attains higher cross-validated accuracy. This suggests that while firm size remains an important determinant of CEO compensation, firm performance adds valuable information that improves the model’s ability to classify pay outcomes.

The optimal number of neighbours is substantially smaller in Model 2 than in Model 1, indicating that the inclusion of firm performance variables makes local neighbourhoods in the feature space more informative. By enriching the feature space, distances between observations better reflect similarities in CEO compensation, allowing accurate predictions to be obtained using fewer neighbours. In contrast, the model based only on firm size requires a larger number of neighbours to stabilise predictions, reflecting the more limited discriminatory power of firm size alone. The higher test accuracy achieved by Model 2 at its optimal value of k further supports the conclusion that firm performance contributes meaningful predictive information.

Moreover, Model 2 also achieves higher test accuracy at its optimal value of k , providing further evidence that the inclusion of firm performance contributes meaningful predictive information. Overall, these results indicate that incorporating firm performance other than firm size makes the feature space more informative and enhances the model’s ability to distinguish between high and low CEO pay.

4.2.2 KNN in regression setting

Followingly, we apply the k -nearest neighbours algorithm in a regression setting to further examine how different measures of firm performance affect the prediction of CEO salary. To obtain an unbiased assessment of model performance and avoid optimistic bias from tuning and evaluation on the same data, we employ a nested K -fold cross-validation procedure. This approach is applied to the same set of model specifications reported in Table 7, including a baseline model focusing on firm size, models emphasising firm performance, and models that jointly incorporate firm size, firm performance, and firm type. For comparison, we also report test MSE from linear regression estimated using standard K -fold cross-validation.

Model	LS Test MSE	KNN Test MSE	KNN Tuned k
Model 1	0.3145	0.2776	7
Model 2	0.3121	0.2552	11
Model 3	0.3123	0.2645	9
Model 4	0.3148	0.2703	9
Model 5	0.3039	0.2595	9
Model 6	0.3025	0.2410	8
Model 7	0.3068	0.2776	6
Model 8	0.4039	0.2964	16

Table 11: Out-of-sample predictive performance of LS and KNN models across different specifications.

Table 11 compares the predictive performance of linear regression (LS) and k -nearest neighbours (KNN) across models that progressively incorporate firm performance measures.

We begin with Model 1, which includes firm size and firm types. Even in this baseline, KNN achieves lower test MSE than LS, suggesting that CEO compensation exhibits local nonlinear patterns that a global linear quadratic function cannot fully capture, although firm size alone already provides substantial predictive power. Models 2–4 add standard performance measures such as *roe*, *pcroe*, and *ros*. The test MSE of LS remains stable around 0.31, indicating that symmetric performance measures add little incremental predictive value beyond firm size. KNN shows modest improvement, reflecting its ability to exploit local variations that LS cannot. A clearer pattern emerges in Models 5–7,

where performance is modelled asymmetrically using a NegROS indicator. Comparing Models 3 and 5, the test MSE of LS declines slightly, whereas KNN improves more substantially. The strongest performance occurs in Model 6, which combines `roe`, `pcroe`, and NegROS. KNN achieves the lowest test MSE = 0.2410, with a small neighbourhood size $k = 8$, indicating that local patterns become more informative when poor performance is explicitly accounted for. Model 8, which excludes firm size, performs poorly under both methods, especially LS. This confirms that performance measures alone are insufficient for accurate prediction and underscores the dominant role of firm size.

Overall, the results indicate that while firm size provides substantial predictive power, KNN captures additional predictive gains by exploiting local and asymmetric patterns in firm performance, particularly negative performance, that linear regression averages out. Even after allowing CEO pay to depend nonlinearly on firm size through a quadratic term, global linear regression still performs slightly worse than KNN in both regression and classification settings. KNN achieves lower out-of-sample MSE and higher classification accuracy by capturing locally heterogeneous relationships between firm size, performance, and pay, as reflected in smaller optimal neighbourhood sizes. In contrast, OLS smooths these effects across the sample, demonstrating the value of modelling local structure in CEO compensation.

4.2.3 Further explanation on KNN predictive performance

The lower test MSE obtained by the KNN model can be explained by its ability to capture non-linear relationships between explanatory variables, in particular `ros` and `lsalesc`. KNN is a non-parametric method, therefore it adapts more flexibly to complex patterns. This non-linear effect is especially relevant when the relationship between firm performance and sales is heterogeneous across observations. Evidence of such non-linearity can be illustrated using partial dependence plots (PDPs), which show that the marginal effects of `ros` and `lsalesc` on the average predicted outcome are not linear.



Figure 6: Partial Dependence Plot of `lsalesc` and ROS

4.3 Decision Tree

4.3.1 Model Specification and Research Design

For our decision trees, we have tested two models, as presented in Table 12. In our research, we have implemented two types of decision trees: a standard decision tree and a redefined one. Our standard decision tree only used GridSearchCV and did not shuffle the data, while our refined decision tree used 5-fold cross-validation and shuffled the data. This shuffling was necessary because our data is ordered by industry sector; without it, the cross-validation folds would have been dominated by single industries, leading to biased results.

Model	Specification
1	$lsalary \sim lsales_c + lsales_c^2 + roe + finance + consprod + utility$
2	$lsalary \sim lsales_c + lsales_c^2 + NegROS + finance + consprod + utility$

Table 12: Regression model specifications for CEO compensation

4.3.2 Standard Decision Tree Analysis and the "Block Trap"

A pivotal finding in this research was the "Block Trap" in the standard version of Model 1. Initial attempts at automated pruning resulted in an underfit, single-node tree that offered zero graphical visualisation or predictive nuance. This was likely a consequence of the lack of explicit shuffling; without it, unbalanced cross-validation folds may have penalised

splitting too heavily. Because decision trees utilise a greedy algorithm, they can easily mistake complex but valid structural patterns for random noise. In this case, the automated pruning logic preferred the safety of a null model over the risk of complexity.

To gain deeper insights, we extended our methodology for Model 1 by creating a second version where we forced the algorithm to find a meaningful split, removing the single-block option from the parameter grid. By forcing this split, we found a 7-node tree and reduced the MSE from 0.3383 to 0.3268, proving that the algorithm’s initial "conservatism" was actually ignoring meaningful relationships. In contrast, Model 2 did not fall into the "Block Trap." Forcing a split resulted in the same 9-node tree, so the standard iteration was already optimal. A comparative analysis reveals significant structural overlap between the two models: both identified *lsales_c* as the primary root node. The divergence occurs exclusively in the left-hand branch (smaller firms): in Model 1, the left branch partitions once based on ROE before terminating, while the left branch in Model 2 exhibits more complexity, splitting on the utility sector indicator followed by log-sales. (See appendix for illustration of the trees (Figure 9))

4.3.3 Refined Decision Tree and the Impact of Constraints

The transition to our Refined Tree initially encountered the same "Block-Trap" for Model 1. In contrast, the Refined Tree for Model 2 yielded a 13-node tree without additional intervention.

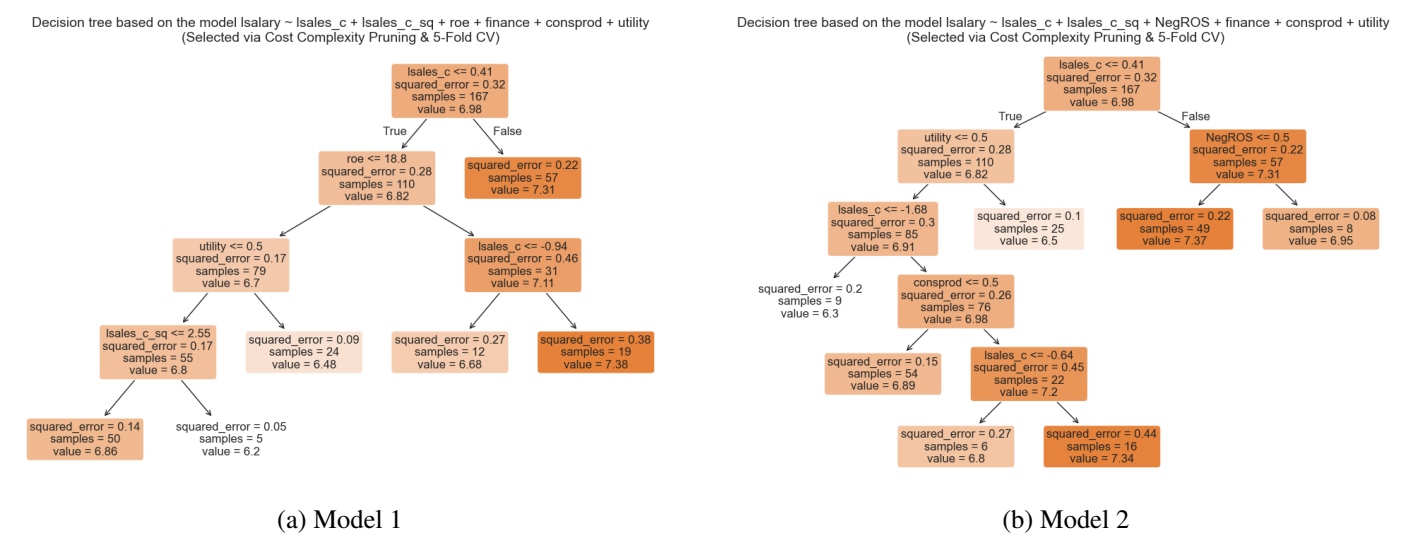


Figure 7: Refined Decision Tree Analysis

The transition to our Refined Tree initially encountered the same "Block-Trap" for Model 1, as the pruning algorithm, favouring simplicity in the face of noise, defaulted once again to a single-node tree. In contrast, the Refined Tree for Model 2 yielded a 13-node tree without additional intervention. To address the persistent underfitting in Model 1 and ensure consistency, we tested two versions for both models: one with a specific constraint on the minimum samples per leaf and one without.

By requiring a minimum number of observations in each leaf, we ensured that each prediction was based on a representative group of CEOs rather than a single outlier, thereby improving the model’s stability and reducing the risk of capturing stochastic noise. For Model 1, this constraint proved transformative: it allowed the model to finally escape the "Block-Trap," expanding from a single node into a nuanced 11-node tree.

This structural expansion was accompanied by a significant gain in predictive power; the tree with the leaf restriction obtained the lowest Mean Squared Error (MSE) of 0.2767 for Model 1 and 0.1924 for Model 2. In contrast, the refined tree without this constraint yielded a significantly higher MSE and, in the case of Model 1, failed to provide any graphical insights. This substantial reduction in error indicates that the unconstrained models were previously suffering from a mix of overfitting (in Model 2) and "conservative" underfitting (in Model 1). By implementing this constraint, we successfully forced the algorithm to prioritise stable, generalizable patterns.

MSE	Model 1 (ROE)	Model 2 (NegROS)
with constraint	0.2767	0.1924
without constraint	0.3383	0.2127

Table 13: MSE without and without minimum samples per leaf

4.3.4 Structural Insights and Branch Logic

Both refined structures reveal an asymmetric hierarchy where the root split is driven by Centered Log-Sales (lsales_c). This confirms that firm size is not merely one predictor among many, but the primary structural driver. In the left branch (Small-to-Mid firms), Model 1 segments by roe immediately following the size split, suggesting that "Pay-for-Performance" is more strictly applied to smaller firms. Both models isolate the Utility sector as reaching a "regulatory ceiling." Furthermore, deeper in this branch, the models incorporate the squared sales term: lsales_c^2 . This indicates Non-Linear Scaling: for mid-market firms, the relationship between size and pay is non-linear, and the model requires this squared term to account for the fact that salary increases at a different rate as a smaller company grows. In the right branch (Large Firms), Model 1 terminates immediately, suggesting a "Negotiation Gap" where CEO pay is highly individualistic. Model 2 offers more nuance here, introducing a split based on NegROS , suggesting that for large-scale firms, compensation is sensitive to whether the firm is profitable or loss-making, rather than the specific degree of performance.

4.3.5 Model Performance and Economic Significance

Beyond the individual tree structures, we implemented a Random Forest to determine if an ensemble approach could better capture the underlying signals in the CEO compensation data. The Random Forest for Model 1 achieved a slightly lower Test MSE as expected, but in Model 2, it yielded an unexpected increase in MSE. This likely stems from our dataset size; with only 209 observations, the dataset may be too sparse to support an ensemble, as the bootstrapping process can lead to training folds that are not representative. This result highlights a critical limitation of ensemble methods: while they are powerful tools for large-scale data, their reliance on diversity through sampling can become a liability when the underlying data observations are scarce. Ultimately, Models 1 and 2 achieved relatively low R^2 . While these look low, they show that while factors like company size and performance are the main drivers, they only explain a small part of the variation. Roughly 87% remains unexplained, suggesting that a large portion of a CEO's salary comes from things our data cannot see, such as unique reputation, specific skills, or private negotiations that require more than just financial data to fully understand.

4.3.6 Random Forest and Feature Importance

Besides predictive power, the Random Forest provided insight into the primary drivers of executive pay, as detailed in Table 14. Our results for Model 1 show that firm size is the undisputed driving factor, with company sales accounting for more than half of the model's predictive weight. The performance of the company also significantly affects compensation, as ROE makes up a substantial portion of the logic. Finally, industry-specific factors account for the remaining weight. In Model 2, the reliance on firm size becomes even more pronounced, accounting for a staggering majority of the model's logic. In this iteration, the performance metric (NegROS) accounts for a much smaller fraction, while industry sectors represent a consistent share.

The divergence between these two models provides critical insight into how compensation is structured. In both models, firm size is the most important feature, which supports the idea that larger firms command higher salaries due to the complexity of the role and the resources at the CEO's disposal. However, the drop in importance from ROE in Model 1 to NegROS in Model 2 is telling of the model mechanics. It suggests that while ROE is a nuanced and powerful predictor, the presence of negative returns is a much weaker signal. This is likely because NegRos is a dummy variable, which limits its impact on splits in non-linear models like a random forest, causing it to underperform compared to continuous variables like ROE. This indicates that while NegRos has an important function in linear models, the random forest prioritises specific, continuous profitability ratios when rewarding CEOs rather than just the binary of whether a firm made a profit or loss. Finally, the fact that industry importance remains relatively stable across both models confirms that sector-specific ceilings are a consistent reality in the CEO labour market.

Feature	Importance	Feature	Importance
$\text{lsales}_c + \text{lsales}_c\text{-sq}$	61.0%	$\text{lsales}_c + \text{lsales}_c\text{-sq}$	82.0%
ROE	26.0%	NegROS	3.0%
Industry Controls	13.0%	Industry Controls	15.0%
R^2	0.1343	R^2	0.0933
MSE	0.2693	MSE	0.2821

Model 1
Model 2

Table 14: Random Forest Comparison: Importance and Diagnostics

4.4 Simulation Study

When we perform our simulation study, in order to test the predictive performance of our different methods used, we found some interesting results.

For the normal simulation without any outliers, we see that the difference in performance between OLS and KNN is very small. Paired t-tests indicate that the mean test MSE of OLS is statistically significantly lower than both KNN and the decision tree. However, while this difference is statistically significant due to the repeated Monte Carlo simulations, the practical difference with KNN is minimal (≈ 0.06 in MSE), meaning that in real-world prediction, the two methods perform almost equally. In contrast, the decision tree shows both a statistically and practically larger difference in MSE (≈ 0.42), suggesting it is less suitable for this type of data. These results are consistent with the relatively small dataset (209 datapoints), where linear relationships are easier to capture with OLS, and KNN only slightly deviates due to the small sample size and the need for scaling. The larger differences with decision trees may be explained by overfitting in small samples and sensitivity to noise in the continuous variables.

Scenario	OLS	KNN (scaled)	KNN (unscaled)	Decision Tree
Normal, no outliers	2.848	2.976	3.027	3.831
Mild outliers (5%)	5.642	5.885	5.892	7.515
Extreme outliers (15%)	39.653	41.207	41.371	49.035

Table 15: Mean Test MSE across 500 Monte Carlo simulations for different models and outlier scenarios

Given that our simulated datasets contain only 209 observations, which is relatively small, the variability of more flexible, nonparametric methods like KNN and decision trees is higher. This means that while their average performance may appear close to OLS, the risk of overfitting or underfitting is more pronounced, especially for models that rely heavily on local patterns in the data. In contrast, OLS, being parametric, benefits from its simplicity and lower variance, making it more reliable in small samples.

For the mild outliers scenario, the predictive performance of all models deteriorates compared to the no-outliers case. The number of outliers visible in the Test MSE increases in Figure 8, in comparison with the scenario without outliers. The presence of a few mild outliers increases the variability in predictions and magnifies the differences between models. In particular, the decision tree shows a larger increase in test MSE relative to the other models, while OLS remains the most stable with slightly lower errors than KNN. The gap between OLS and KNN increases slightly compared to the no-outliers scenario, indicating that KNN is slightly more affected by these mild deviations.

In the extreme outliers scenario, predictive performance declines drastically for all models. There are an extreme number of Test MSE outliers visible in Figure 8. Extreme observations strongly influence predictions, and the differences between models become more pronounced. The decision tree suffers the most, whereas OLS maintains relatively lower errors, outperforming both KNN and the tree by a notable margin. The gap in test MSE between OLS and KNN is now larger than in the mild outliers scenario, reflecting the higher sensitivity of flexible, nonparametric models to extreme values.

These results illustrate a key difference between parametric and nonparametric approaches in the presence of outliers. When no outliers are present, KNN and OLS produce very similar predictive performance, with mean test MSEs close to each other, indicating that both methods can capture the underlying data structure effectively. As outliers are introduced and their severity increases, the performance of KNN begins to diverge from OLS, showing higher sensitivity to extreme observations.

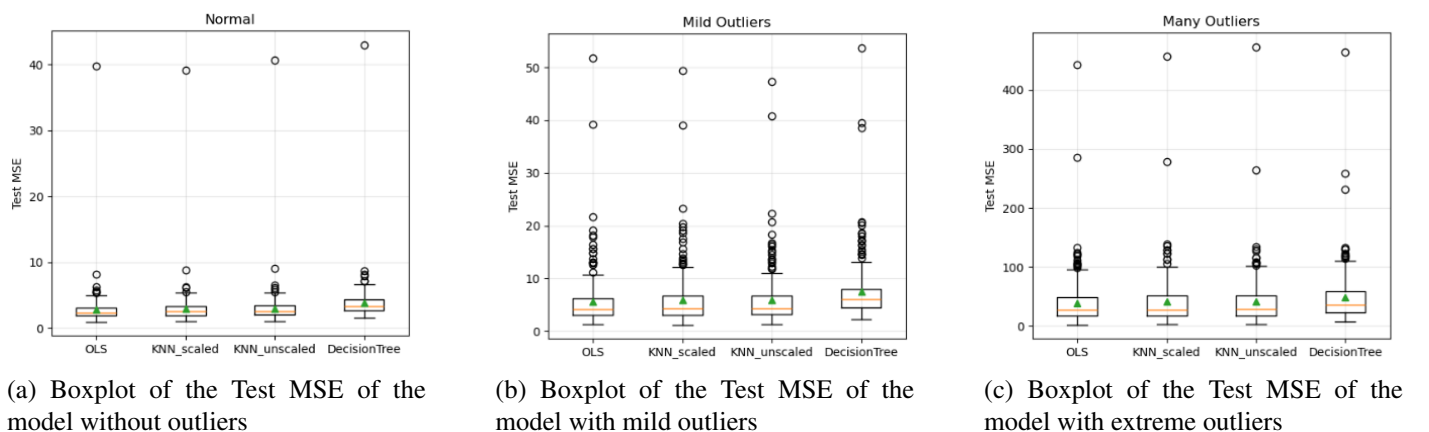


Figure 8: Monte Carlo simulation: Test MSE per DGP.

Decision trees are particularly affected by outliers. The splitting rules in a tree are based directly on the observed values, and extreme values can influence which splits are chosen. As a result, the tree may focus on fitting these outliers rather than the general trend of the data. This sensitivity explains why decision trees show the largest increase in test MSE in the presence of outliers, especially when extreme values are present. In contrast, OLS remains more stable because its linear parametric structure is less influenced by individual extreme observations. Overall, the results show that while KNN and OLS perform similarly in clean datasets, outliers increasingly amplify the differences in predictive performance, and decision trees are the most impacted.

It is also notable that the difference between KNN with scaled and unscaled predictors is minimal across all scenarios. The mean test MSEs differ only very slightly (less than 0.01 in all cases), suggesting that the continuous predictors do not differ substantially in scale, and that scaling has little effect on KNN predictions in this particular simulation.

In conclusion, OLS remains the best-performing model across all scenarios, despite its predictive performance deteriorating in the presence of outliers. KNN performs similarly to OLS when no outliers are present, but becomes increasingly sensitive as outliers are introduced, while decision trees are the most affected due to their reliance on observed values for splits. Scaling has minimal impact on KNN, reflecting the relatively similar scale of the continuous predictors in this DGP.

5 Conclusion

5.1 Main procedure

This study investigated the determinants of CEO compensation and analysed whether executive pay is primarily driven by firm performance measures such as return on equity and stock returns. It explored how various firm characteristics and performance metrics are utilised to determine compensation. The analysis was based on a dataset of 209 observations, examining variables such as sales, return on equity, and industry-specific factors.

The methodological procedure began with the estimation of a simple linear regression as a starting point, which was then expanded into a more realistic model by incorporating log-level variables, industry dummies, and non-linear relationships to better capture the complexities of the data. The analysis further extended to non-parametric methods, utilising K-Nearest Neighbours and Decision Trees to address potential non-linearities without assuming a fixed functional form. To evaluate and compare the predictive performance of these models, k-fold and Leave-One-Out cross-validation were employed, alongside ensemble methods like Random Forests to enhance stability and reduce variance. Finally, a Monte Carlo simulation study was conducted to assess the robustness of the findings across different scenarios, specifically focusing on the impact of mild and extreme outliers on each model's predictive accuracy.

5.2 Findings

From an inferential standpoint, firm size emerges as the dominant determinant of CEO compensation across all model specifications, while firm performance has a statistically detectable but economically small effect. The simple linear regression already explains a substantial share of the variation in CEO salary, and the inclusion of firm performance measures leads only to a modest increase in adjusted- R^2 . Among the performance variables considered, the continuous measure return on equity (`roe`) provides the most consistent additional explanatory power. Other performance measures, such as `ros` and `pcroe`, are statistically insignificant and do not meaningfully improve model fit. Interestingly, the qualitative indicator `NegROS`, capturing whether firm performance is negative, performs even slightly better than `roe` in explaining the variation in CEO pay, suggesting that compensation may respond more strongly to threshold effects like losses.

From a predictive perspective, firm size remains the dominant determinant of CEO compensation across all methods. All three methods confirm this. In CV and KNN, the baseline model highlighting firm size achieves relatively low cross-validated MSE and test MSE; decision tree results reinforce these findings, with firm size constituting the primary split before any other characteristics. Linear regression shows that adding standard performance measures beyond firm size provides minimal predictive gains, although including the asymmetric negative performance indicator (`NegROS`) slightly improves cross-validated MSE. KNN shows the same results with better achievement on using (`NegROS`) in predicting salary. The decision tree results further illustrate this heterogeneity: CEO pay in large firms is mainly driven by firm size, whereas industry effects play a larger role for small- to mid-sized firms. Firm performance appears as a secondary predictor, typically influencing splits only after size thresholds are determined. Overall, the three methods collectively suggest that while global parametric models can capture the dominant role of size, nonparametric approaches like KNN and decision trees reveal local structure and asymmetric performance effects, which substantially improve predictive accuracy.

The simulation study highlights clear trade-offs between the methods. In clean datasets, KNN and OLS achieve nearly identical predictive performance. However, when outliers are introduced, OLS proves more robust, as its parametric structure offers greater stability in small samples. In contrast, flexible methods such as KNN and decision trees deteriorate as outlier severity increases. While these models are well suited to capturing non-linearities and local patterns, the simplicity and stability of OLS make it the most reliable choice under extreme observations and limited data.

From both an inferential and predictive standpoint, our findings suggest that while performance is a statistically relevant factor, its influence is secondary to the structural dominance of firm size. The inclusion of performance measures like `roe` and `NegROS` provides only modest improvements in explanatory and predictive power compared to the overwhelming impact of firm scale. Crucially, models constructed solely on firm performance variables actually performed worse, demonstrating that performance metrics lack a sufficient independent signal to explain compensation without the context of firm size. Ultimately, CEO pay is anchored primarily by the size and complexity of the firm, with performance metrics acting as a necessary secondary tool that must be combined with firm scale to achieve any meaningful level of accuracy and explanatory depth.

5.3 Future research

Despite the depth of the analysis, the study is not without limitations point to directions for future research. A primary constraint is the reliance on a cross-sectional dataset from a single year, which prevents the analysis of how CEO compensation evolves over time in response to long-term firm performance.

Furthermore, the measurement of firm size presents a significant limitation. Firm size is currently captured by sales, which may not fully reflect other dimensions such as market value, total assets, or the number of employees. It may also be endogenous to CEO pay, as better-paid CEOs could influence firm growth. Future research should therefore consider alternative size measures or causal methods to better isolate these relationships. The relatively small sample size of 209 observations also suggests that results could be refined with a larger dataset to improve generalisability.

Moreover, the reported R^2 and adjusted- R^2 values indicate that a significant portion of the variation in CEO salary remains unexplained. This highlights the need to explore other variables, such as CEO tenure, education, gender, or board composition, which might play a significant role in the bargaining power of executives. Besides, exploring more nuanced performance metrics beyond `roe` and `ros` could provide a better understanding of how firms determine executive compensation in modern corporate environments.

Finally, it is important to note that the dataset used in this study is relatively outdated, covering firms from 1990. The relationship between firm size, firm performance, and CEO compensation may have evolved substantially since then. More recent and larger-scale studies suggest a stronger link between performance and CEO pay, and also highlight how major shocks, such as the COVID-19 pandemic, have reshaped compensation structures and incentive schemes. (Robin Klaver, 2024) As a result, the findings in this report should be interpreted in light of these changing economic conditions.

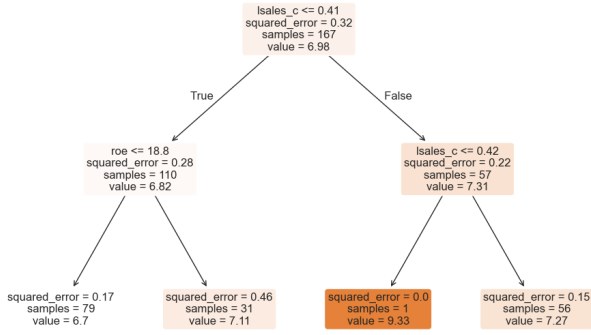
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6 Appendix

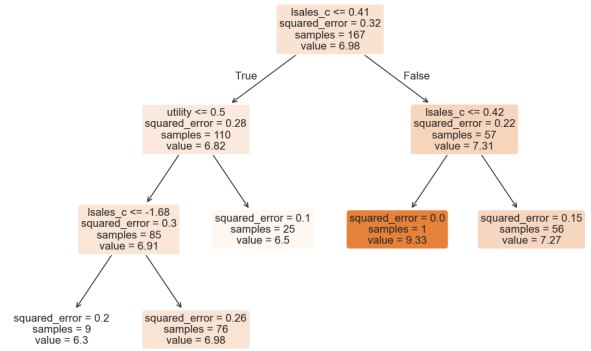
Figure 9: Illustration Standard decision tree

Standard Decision tree based on the model $\text{Isalary} \sim \text{Isales_c} + \text{Isales_c_sq} + \text{roe} + \text{finance} + \text{consprod} + \text{utility}$ (without shuffled data)



(a) Model 1

Standard Decision tree based on the model $\text{Isalary} \sim \text{Isales_c} + \text{Isales_c_sq} + \text{NegROS} + \text{finance} + \text{consprod} + \text{utility}$ (without shuffled data)



(b) Model 2

Figure 9: Standard Decision Tree Analysis

6.0.1 Testing for assumptions

The validity of ordinary least squares regression relies on several key assumptions. To ensure reliable inference and robust estimation, these assumptions are formally tested.

Homoscedasticity, or constant variance of the error terms, is examined using the Breusch–Pagan test. A failure to reject the null hypothesis indicates that the variance of the residuals does not systematically vary with the explanatory variables, supporting the use of conventional standard errors.

Multicollinearity among predictors is assessed through Variance Inflation Factors (VIFs). High VIF values indicate strong correlations between regressors, which can inflate standard errors and complicate the interpretation of individual coefficients, even when the overall model remains predictive.

Functional form misspecification is evaluated using the RESET test, which tests whether nonlinear combinations of the fitted values improve model fit. Non-significant results provide evidence that the linear specification adequately captures the relationships in the data. Independence of residuals is assessed using the Durbin–Watson statistic, ensuring that residuals are not serially correlated. For the Durbin–Watson test, values close to 2 indicate no autocorrelation between the residuals.

Normality of residuals is evaluated with the Jarque–Bera test to verify that skewness and kurtosis are consistent with a normal distribution, meaning we have normal errors when the test does not reject the null hypothesis.

Meeting these assumptions ensures that standard errors, confidence intervals and hypothesis tests are valid and that coefficient estimates are interpretable. Collectively, these diagnostic tests provide a comprehensive evaluation of whether the classical assumptions of linear regression are satisfied, supporting both the validity of inference and the reliability of predictive performance.